Work, energy and power CHEAT SHEET

WORK

$W = F\,d\cos\theta$ (only the part along the motion works)

unit: 1 J = 1 N × 1 m (a 100 g apple lifted 1 m ≈ 1 J)

sledge: 60 N at 60° for 15 m → 30 N along → 450 J

edges: $\theta = 0°$ all of F; 90° none; > 90° negative

crate: push +720 J, friction −540 J, weight 0 → net 180 J

backwards: $\cos\theta = \frac{W}{F\,d} = \frac{450}{900} \rightarrow \theta = 60°$

VARIABLE FORCES

work = area under F–x = $\int F\,dx$

30 N → 10 N over 4 m: 40 + 40 = 80 J

5 kg trolley, simulated: 5.66 m/s → $\frac{1}{2}mv^2 = 80$ J

$F = m\,v\,\frac{dv}{dx} \rightarrow \int F\,dx = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$

constant F: the area is a rectangle, F d

KINETIC ENERGY

steady a: $v^2 = u^2 + 2ad$; × m/2: $F\,d = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$

kinetic energy $K = \frac{1}{2}\,m\,v^2$ (never negative)

football 0.45 kg at 20 m/s → 90 J

twice the speed, four times the energy: car 200 → 800 kJ

work-energy theorem: $W_{net} = \Delta K$

braking 7,000 N, 1,400 kg: 15 m/s → 22.5 m; 30 m/s → 90 m

backwards: skid 40 m at 5,000 N → $v = \sqrt{(2Fd/m)} \approx 16.9$ m/s

POTENTIAL ENERGY

$U = m\,g\,h$ (from a level you choose)

2 kg bag onto a 1.5 m shelf: 29.4 J

gravity's work: height only, not path (conservative)

4 m ramps, straight and curved: both 8.85 m/s

spring: $F = -k\,x$; $U = \frac{1}{2}\,k\,x^2$ (the triangle's area)

200 N/m squeezed 0.1 m → 1 J; 0.2 m → 4 J

0.5 kg launched, simulated: 2 m/s

THE ENERGY ACCOUNT

conservative only: K + U stays the same

phone off 0.8 m: $v = \sqrt{(2gh)} = 3.96$ m/s, any mass

$W_{nc} = \Delta K + \Delta U$ (friction's work leaves as heat)

3 kg at 4 m/s, $\mu = 0.25$: stops in 3.27 m, 24 J heat

slide 2.5 m: 7 m/s ideal; 5 m/s real → 300 J, 60 N

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THE VERTICAL CIRCLE

inward pull for a circle: $\frac{m\,v^2}{R}$

top: $T + mg = \frac{m\,v^2}{R}$, $T \geq 0 \rightarrow v_{top}^{\,2} \geq g\,R$

bottom: $v^2 = v_{top}^{\,2} + 4gR \geq 5gR \rightarrow v \geq \sqrt{(5gR)}$

R = 1 m: 3.13 m/s at the top, 7.00 m/s at the bottom

string at the bottom: $T = 6\,m\,g$ (2 kg → 117.6 N)

simulated: T = 0 exactly at the top

rod: $\sqrt{(4gR)} = 6.26$ m/s; a string then goes slack at 5R/3

ramp into a loop: start ≥ 2.5 R high

POWER

$P = \frac{W}{t} = F\,v$; 1 W = 1 J/s; 1 kWh = 3.6 MJ

lift 800 kg up 20 m in 16 s: 156,800 J → 9.8 kW

$F\,v = 7{,}840\text{ N} \times 1.25\text{ m/s} = 9{,}800\text{ W}$

cyclist 25 N at 8 m/s: 200 W; car 30 kW at 25 m/s: 1,200 N

1 horsepower = 33,000 ft·lbf/min ≈ 745.7 W

COLLISIONS

p = m v: kept in every collision

elastic: K kept too; inelastic: some K lost

stick: $v = \frac{m_1u_1 + m_2u_2}{m_1 + m_2}$ (30 t at 2 m/s + 20 t → 1.2 m/s)

60 kJ → 36 kJ: 40 % lost (equal masses: half)

elastic, $u_2 = 0$: $v_1 = \frac{m_1 - m_2}{m_1 + m_2}u_1$, $v_2 = \frac{2m_1}{m_1 + m_2}u_1$

2 kg at 3 m/s on 1 kg → 1 and 4 m/s (simulated)

edges: equal → swap; heavy on light → ≈ 2u; light on heavy → −u

TWO DIMENSIONS

2-D: p kept east and north separately

80 kg east 6 m/s + 100 kg north 4 m/s → 3.47 m/s at 39.8°

equal balls, elastic glance: 90° apart (2 → 1.73 and 1 m/s)

pellet 10 g in 1.99 kg, rises 10 cm → 1.4 m/s → 280 m/s

THE ROLLER COASTER'S LOOP

loop 20 m tall: R = 10 m; launched level; no friction

the seat, like a string, only presses inwards

top: $v_{top}^{\,2} \geq g\,R = 98$

launch: $v \geq \sqrt{(5gR)} = \sqrt{490} = 22.14$ m/s ≈ 80 km/h

at exactly that, riders float over the top

24 m/s: top 13.56 m/s (simulated), seat 0.88 × weight